Supplementary materials

Comparative efficacy and safety of systemic treatments for cutaneous leishmaniasis: a systematic review and network meta-analysis

Mahmood Fadaie[1,2], Sajjad Biglari[1], Anis Khalafiyan[1], Maryam Yazdi[3,4], Hassan Vahidnezhad[5,6,7,8,*], Zabihollah Shahmoradi[2,9], and Hossein Khanahmad[1,2,*]

[1]Department of Genetics and Molecular Biology, School of Medicine, Isfahan University of Medical Sciences, Isfahan, I.R. Iran. [2]Skin Diseases and Leishmaniasis Research Center, Isfahan University of Medical Sciences, Isfahan, I.R. Iran. [3]Child Growth and Development Research Center, Research Institute for Primordial Prevention of Non-Communicable Diseases, Isfahan University of Medical Sciences, Isfahan, I.R. Iran. [4]Diabetes Research Centre, College of Life Sciences, University of Leicester, Leicester, UK. [5]Center for Applied Genomics, Children's Hospital of Philadelphia, Philadelphia, PA, USA. [6]Division of Human Genetics, Children's Hospital of Philadelphia, Philadelphia, PA, USA. [7]Department of Pediatrics, University of Pennsylvania, Perelman School of Medicine, Philadelphia, PA, USA. [8]Department of Dermatology, University of Pennsylvania, Perelman School of Medicine, Philadelphia, PA, USA. [9]Department of Dermatology, Isfahan University of Medical Sciences, Isfahan, I.R. Iran.

[*]Corresponding authors:
H. Khanahmad, Tel/Fax: +98-3137929197
Email: hossein_khanahmad@yahoo.com
H. Vahidnezhad, Tel/Fax: +1-2154299783
Email: vahidnezhh@chop.edu

Appendix S1

Complete search strategies including as follows:

PubMed

((“systemic treatment*”[tiab] OR “systemic therapy”[tiab] OR “pentavalent antimon*”[tiab] OR therapeutics[tiab] OR “drug therapy”[tiab] OR “azole molecules”[tiab] OR “amphotericin B”[tiab] OR miltefosine[tiab] OR “pentamidine isethionate”[tiab] OR paromomycin[tiab] OR immunotherapy[tiab] OR immunomodulat*[tiab] OR GM-CSF[tiab] OR “granulocyte macrophage colony-stimulating factor”[tiab] OR “zinc therapy”[tiab] OR “sodium stibogluconate”[tiab] OR “meglumine antimoniate”[tiab] OR fluconazole[tiab] OR ketoconazole[tiab] OR itraconazole[tiab] OR cyclosporin[tiab] OR “dihydrocyclosporin A”[tiab] OR Imiquimod[tiab])
AND
(“Leishmaniasis, Cutaneous”[mesh] OR “cutaneous leishmaniasis”[tiab] OR “cutaneous leishmaniases”[tiab] OR “leishmaniases, cutaneous”[tiab] OR “leishmaniasis, old world”[tiab] OR “old world leishmaniasis”[tiab] OR “leishmaniasis, new world”[tiab] OR “new world leishmaniasis”[tiab])
AND
(“randomized controlled trial”[mesh] OR “controlled clinical trial”[mesh] OR randomly[tiab] OR randomized[tiab] OR trial[tiab] OR “cohort studies”[tiab]))

Scopus

(TITLE-ABS-KEY ("systemic treatment*") OR TITLE-ABS-KEY ("systemic therapy") OR TITLE-ABS-KEY ("pentavalent antimon*") OR TITLE-ABS-KEY (therapeutics) OR TITLE-ABS-KEY ("drug therapy") OR TITLE-ABS-KEY ("azole molecules") OR TITLE-ABS-KEY ("amphotericin B") OR TITLE-ABS-KEY (miltefosine) OR TITLE-ABS-KEY ("pentamidine isethionate") OR TITLE-ABS-KEY (paromomycin) OR TITLE-ABS-KEY (immunotherapy) OR TITLE-ABS-KEY (immunomodulat*) OR TITLE-ABS-KEY (gm-csf) OR TITLE-ABS-KEY ("granulocyte macrophage colony-stimulating factor") OR TITLE-ABS-KEY ("zinc therapy") OR TITLE-ABS-KEY ("sodium stibogluconate") OR TITLE-ABS-KEY ("meglumine antimoniate") OR TITLE-ABS-KEY (fluconazole) OR TITLE-ABS-KEY (ketoconazole) OR TITLE-ABS-KEY (itraconazole) OR TITLE-ABS-

KEY (cyclosporin) OR TITLE-ABS-KEY ("dihydrocyclosporin A") OR TITLE-ABS-KEY (imiquimod) AND TITLE-ABS-KEY ("Leishmaniasis, Cutaneous") OR TITLE-ABS-KEY ("cutaneous leishmaniasis") OR TITLE-ABS-KEY ("cutaneous leishmaniases") OR TITLE-ABS-KEY ("leishmaniases, cutaneous") OR TITLE-ABS-KEY ("leishmaniasis, old world") OR TITLE-ABS-KEY ("old world leishmaniasis") OR TITLE-ABS-KEY ("leishmaniasis, new world") OR TITLE-ABS-KEY ("new world leishmaniasis") AND TITLE-ABS-KEY ("randomized controlled trial") OR TITLE-ABS-KEY ("controlled clinical trial") OR TITLE-ABS-KEY (randomly) OR TITLE-ABS-KEY (randomized) OR TITLE-ABS-KEY (trial) OR TITLE-ABS-KEY ("cohort studies")) View less

Embase

('systemic treatment*':ti,ab,kw OR 'systemic therapy':ti,ab,kw OR 'pentavalent antimon*':ti,ab,kw OR therapeutics:ti,ab,kw OR 'drug therapy':ti,ab,kw OR 'azole molecules':ti,ab,kw OR 'amphotericin b':ti,ab,kw OR miltefosine:ti,ab,kw OR 'pentamidine isethionate':ti,ab,kw OR paromomycin:ti,ab,kw OR immunotherapy:ti,ab,kw OR immunomodulat*:ti,ab,kw OR 'gm csf':ti,ab,kw OR 'granulocyte macrophage colony-stimulating factor':ti,ab,kw OR 'zinc therapy':ti,ab,kw OR 'sodium stibogluconate':ti,ab,kw OR 'meglumine antimoniate':ti,ab,kw OR fluconazole:ti,ab,kw OR ketoconazole:ti,ab,kw OR itraconazole:ti,ab,kw OR cyclosporin:ti,ab,kw OR 'dihydrocyclosporin a':ti,ab,kw OR imiquimod:ti,ab,kw) AND ('leishmaniasis, cutaneous':ti,ab,kw OR 'cutaneous leishmaniasis':ti,ab,kw OR 'cutaneous leishmaniases':ti,ab,kw OR 'leishmaniases, cutaneous':ti,ab,kw OR 'leishmaniasis, old world':ti,ab,kw OR 'old world leishmaniasis':ti,ab,kw OR 'leishmaniasis, new world':ti,ab,kw OR 'new world leishmaniasis':ti,ab,kw) AND ('randomized controlled trial':ti,ab,kw OR 'controlled clinical trial':ti,ab,kw OR randomly:ti,ab,kw OR randomized:ti,ab,kw OR trial:ti,ab,kw OR 'cohort studies':ti,ab,kw)

Web of Science

((“systemic treatment*” OR “systemic therapy” OR “pentavalent antimon*” OR therapeutics OR “drug therapy” OR “azole molecules” OR “amphotericin B” OR miltefosine OR “pentamidine isethionate” OR paromomycin OR immunotherapy OR immunomodulat* OR GM-CSF OR “granulocyte macrophage colony-stimulating factor” OR “zinc therapy” OR “sodium stibogluconate” OR “meglumine antimoniate” OR fluconazole OR ketoconazole OR itraconazole OR cyclosporin OR “dihydrocyclosporin A” OR Imiquimod)
AND
(“Leishmaniasis, Cutaneous” OR “cutaneous leishmaniasis” OR “cutaneous leishmaniases” OR “leishmaniases, cutaneous” OR “leishmaniasis, old world” OR “old world leishmaniasis” OR “leishmaniasis, new world” OR “new world leishmaniasis”)
AND
(“randomized controlled trial” OR “controlled clinical trial” OR randomly OR randomized OR trial OR “cohort studies”))

Supplementary figures

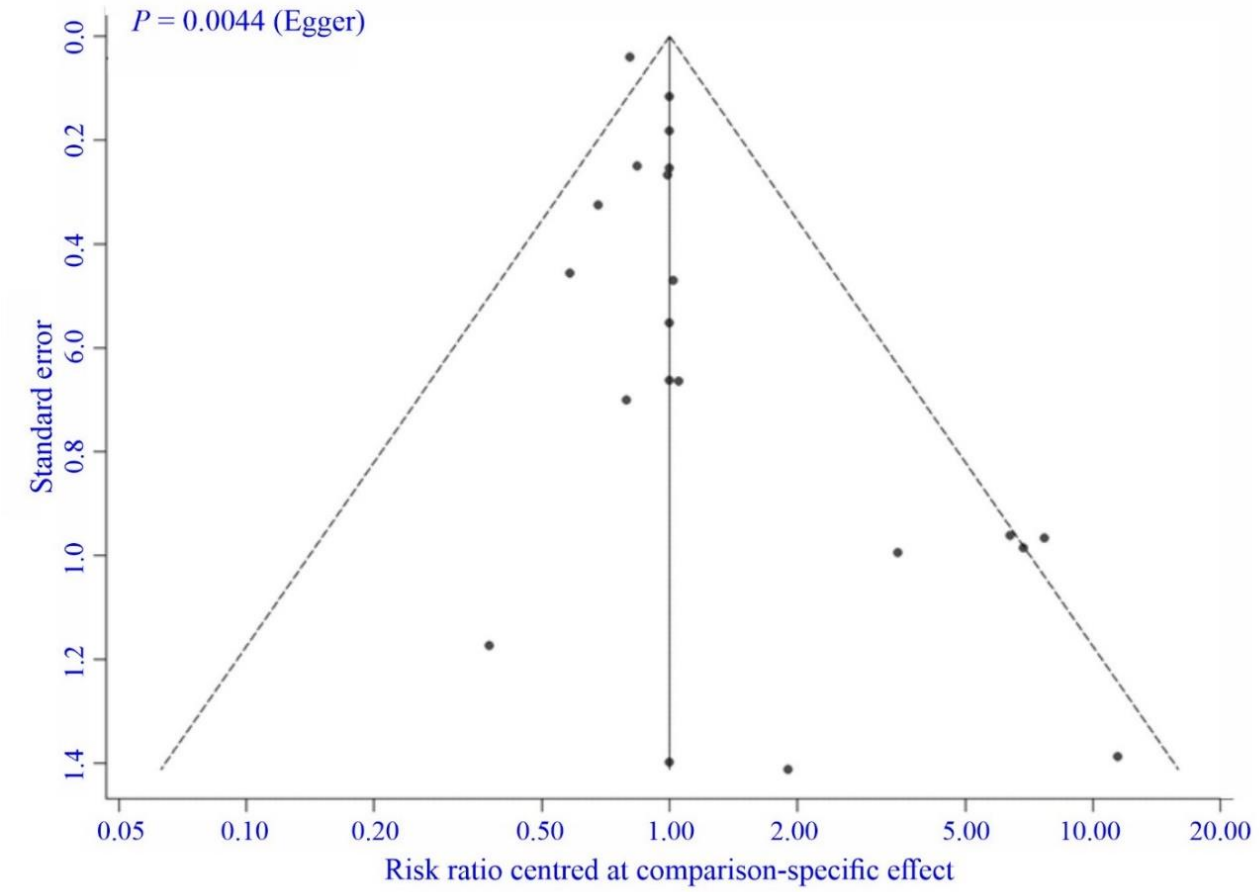


Fig. S1. Comparison-adjusted funnel plot assessing small-study effects in the network meta-analysis excluding high-risk studies.

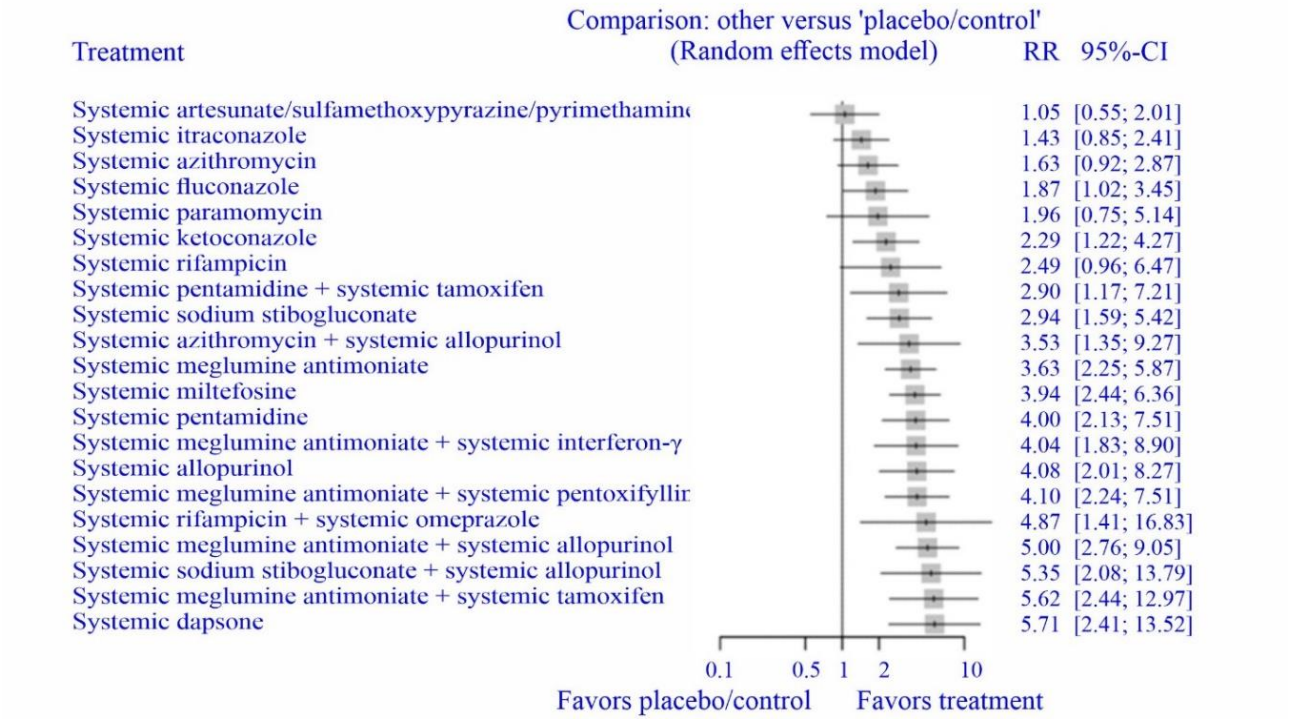


Fig. S2. Forest plot from a network meta-analysis comparing systemic treatments against placebo/control for cutaneous leishmaniasis. RR with 95% CI are shown. The horizontal line denotes the 95% CI, and the vertical line indicates no effect (RR = 1). The I^2 statistic for heterogeneity was 68.8%. RR, Relative risk; CI, confidence interval.

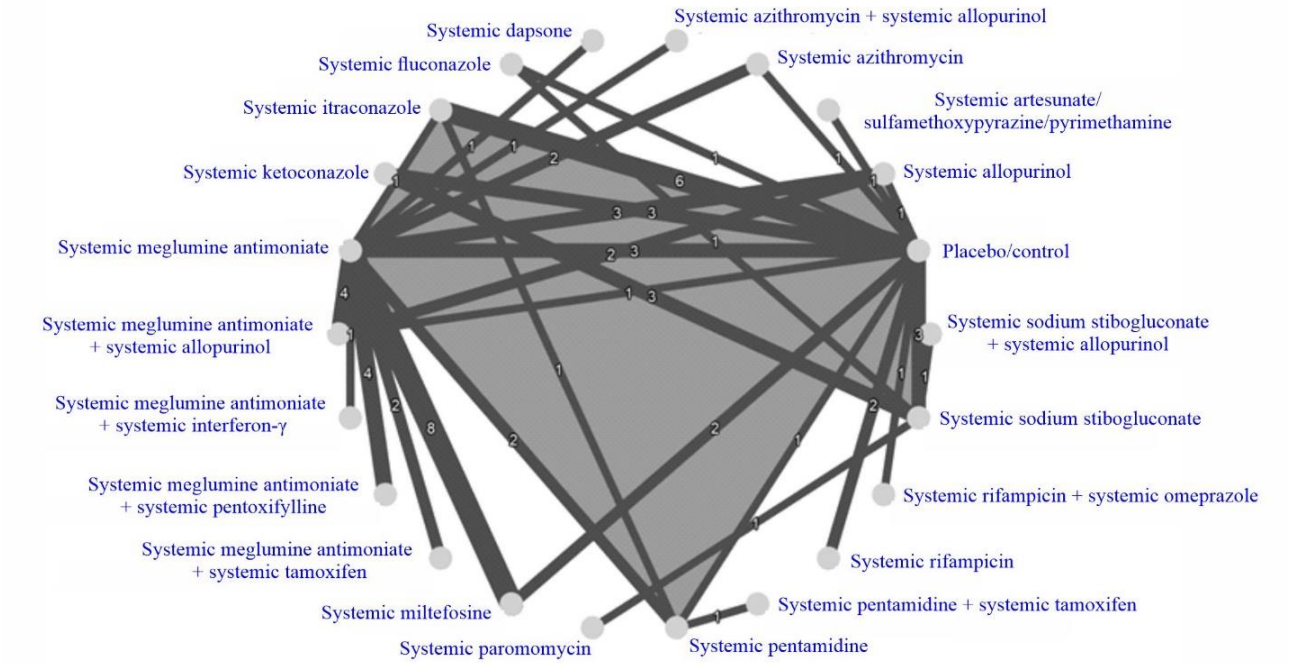


Fig. S3. Network plot of systemic treatments for cutaneous leishmaniasis. Node size represents the number of participants/intervention arms, and line thickness represents the number of direct comparisons between treatments.

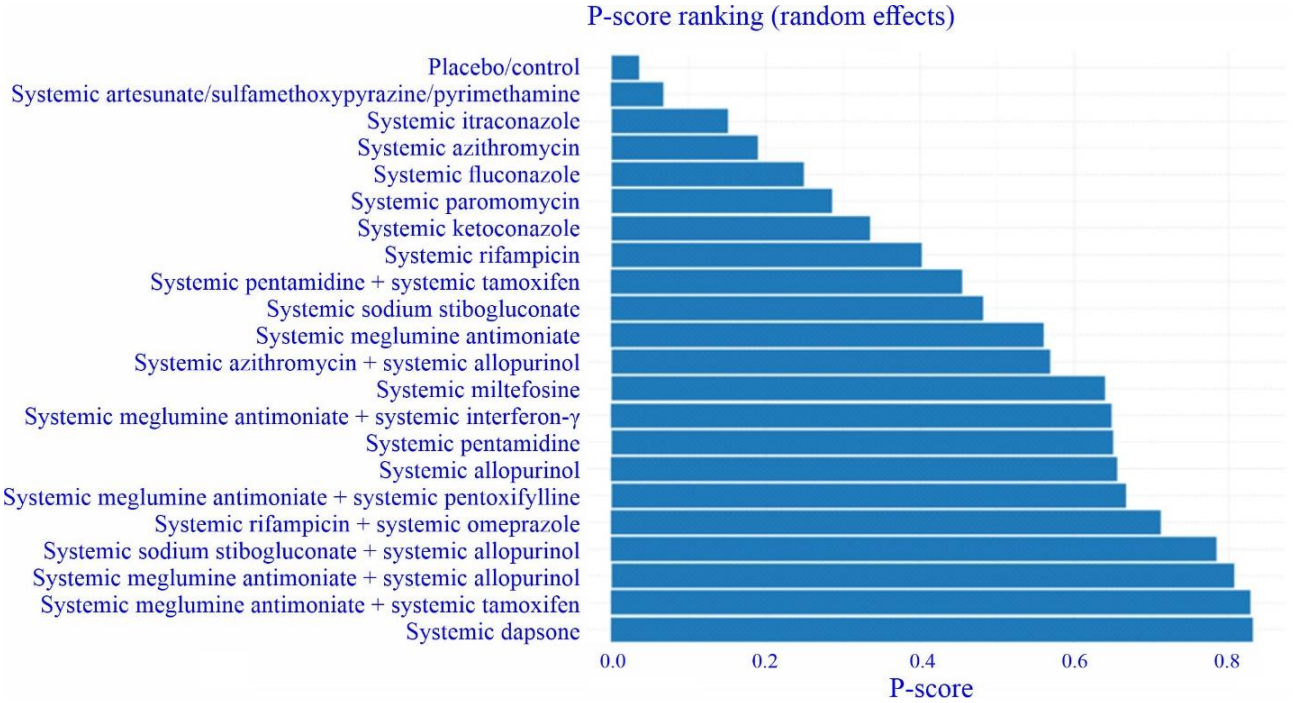


Fig. S4. P-score ranking of systemic treatments for cutaneous leishmaniasis from a network meta-analysis. Higher P-scores indicate greater confidence in a treatment's efficacy. Rankings are shown for the random-effects model. Systemic dapsone achieves the highest rank, followed by combination therapies such as meglumine antimoniate + tamoxifen, sodium stibogluconate + allopurinol, and meglumine antimoniate + allopurinol. Placebo/control consistently ranks lowest.

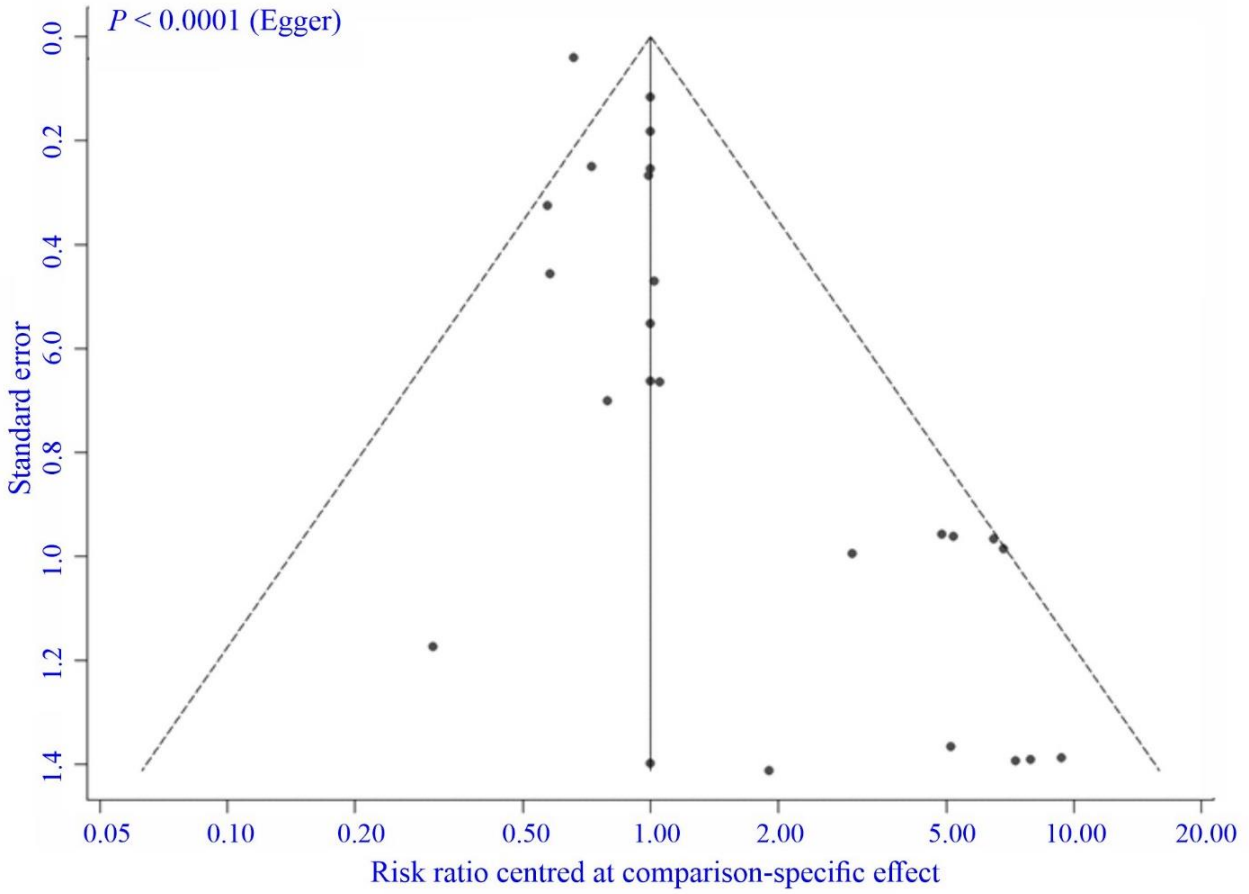


Fig. S5. Comparison-adjusted funnel plot assessing small-study effects in the overall network meta-analysis.

Supplementary tables

Table S1. Baseline characteristics of RCTs included in qualitative analysis.

Study	Age range or mean (year)	*Leishmania* Species	Intervention (number of participants)	Cure rate (% per group)	Follow-up (weeks)	AEs
Abtahi-Naeini *et al.*, 2021 (1)	25.9	*L. major*	Oral azithromycin (14, 20 lesions)	65%	12	No significant AEs
	25.1		Oral + topical liposomal azithromycin (13, 21 lesions)	85.7%		Mild pruritus
Adam *et al.*, 2009 (2)	30.6	Not specified	Oral artesunate + sulfamethoxypyrazine/pyrimethamine (20)	90%	10	No significant AEs
Ahmed *et al.*, 2022 (3)	31 (20-80)	Not specified	IM meglumine antimoniate (50)	42%	11	Four treatment discontinuations due to AE
			Oral dapsone (50)	66%		Five treatment discontinuations due to AE
Alfahaad, 2020 (4)	5-65	Not specified	IM + IL sodium stibogluconate (20 patients)	60%	20	Systemic AEs were absent
Al-Fouzan *et al.*, 1991 (5)	12-52	Not specified	Oral itraconazole (15)	73%	12	Nausea, headache, and elevated liver enzymes
Alkhawajah *et al.*, 1997 (6)	13-42	*L. major*	IM meglumine antimoniate (31, 68 lesions)	68%	4	No significant AEs
Almeida *et al.*, 1999 (7)	16-51	*L. braziliensis*	IV sodium stibogluconate (10)	100%	42	No significant AEs
			IV sodium stibogluconate + IL GM-CSF (10)	100%		Muscle aches and general malaise
Alrajhi *et al.*, 2002 (8)	31.2	*L. major*	Oral fluconazole (80)	79%	12	Mild AEs
Arana *et al.*, 1994 (9)	20.1	*L. braziliensis* and *L. mexicana*	IV meglumine antimoniate for 20 days (21)	90%	52	Mild AEs
	18.9		IV meglumine antimoniate for 10 days (20)	90%		Mild AEs
	19.6		IV meglumine antimoniate + interferon-γ for 10 days (22)	100%		Mild AEs such as mild malaise, headache and fever
Arevalo *et al.*, 2007 (10)	34.9	Not specified	IV meglumine antimoniate (7)	57%	12	Mild AEs
			IV meglumine antimoniate + imiquimod cream (7)	100%		Mild to moderate erythema, edema and sensations of burning
Armijos *et al.*, 2004 (11)	21.3	*Leishmania* from the *Viannia* subgenus	IM meglumine antimoniate (36)	91.7%	52	Systemic AEs such as fever, headache and weakness

Aronson *et al.*, 2010 (12)	18-57	*L. major*	IV sodium stibogluconate (26)	54%	52	Reversible abdominal pain/pancreatitis, arthralgias, myalgias, headache, fatigue, and elevated liver enzymes.
Bashir *et al.*, 2019 (13)	16-60	Not specified	Oral itraconazole + IL meglumine antimoniate (35)	100%	12	No significant AEs
Brito *et al.*, 2014 (14)	34 29	*L. braziliensis*	IV meglumine antimoniate (15) IV meglumine antimoniate + oral pentoxifylline (18)	47% 56%	12	Mild AEs, such as arthralgia, headache, and fever, were more common in the pentoxifylline group
Brito *et al.*, 2017 (15)	18-62	*L. braziliensis*	IV meglumine antimoniate (82) IV meglumine antimoniate + oral pentoxifylline (82)	43% 45%	24	The pentoxifylline group had 37.8% AEs compared to 23% for the other group
Castro *et al.*, 2022 (16)	18-62	*L. panamensis*, *L. braziliensis* and *L. guyanensis*	IM meglumine antimoniate (36) IM meglumine antimoniate + oral pentoxifylline (34)	75% 64.7%	26	Similar AEs, with fever, headache, elevated liver enzymes, and injection site responses being the most common
Celes *et al.*, 2022 (17)	19-42	*L. braziliensis*	IV meglumine antimoniate (9) IV meglumine antimoniate + topical bacterial cellulose bio-curatives (10)	55% 80%	12	No AEs were reported for either group
Chrusciak-Talhari *et al.*, 2011 (18)	4-62	*L. guyanensis*, *L. braziliensis* and *L. lainsoni*	Oral miltefosine (56) IV meglumine antimoniate (28)	71.4% 53.6%	24	48.3% reported vomiting, 8.6% nausea, 6.7% diarrhea, and 3.4% orchitis 33% reported arthralgias
D'Oliveira Jr. *et al.*, 1997 (19)	12-45	*L. braziliensis*	Oral allopurinol (18) IV meglumine antimoniate (16)	0% 50%	52	Four patients in the allopurinol group experienced worsened ulcers, edema, and pain. One patient developed mucosal disease
Dastgheib *et al.*, 2012 (20)	39.7 36.8	*L. major*	Oral azithromycin + allopurinol (36) IM meglumine antimoniate (35)	38.9% 40%	8	Minimal AEs. One patient developed gastrointestinal issues Minimal AEs. two patients experienced myalgia
Dogra *et al.*, 1990 (21)	14-56	*L. major* and *L. tropica*	Oral itraconazole (15)	66.6%	12	Three patients experienced mild headache
Dogra *et al.*, 1994 (22)	15-62	*L. major* and *L. tropica*	Oral itraconazole (20)	75%	24	Three patients experienced transient liver function test abnormalities
Dogra *et al.*, 1996 (23)	19-62	*L. major* and *L. tropica*	Oral itraconazole (10)	80%	12	One patient experienced elevated liver enzymes and

						another patient reported nausea
El-Sayed *et al.*, 2010 (24)	12-50	Not specified	IM + IL sodium stibogluconate (10, 15 lesions) Oral ketoconazole + IL sodium stibogluconate (10, 13 lesions)	93.3% 92.3%	24	No systemic AEs were observed
Esfandiarpour *et al.*, 2002 (25)	14	Not specified	Oral allopurinol (50) IM meglumine antimoniate (50) Oral allopurinol + IM meglumine antimoniate (50)	18% 24% 46%	4	Nausea, heartburn, and mild increases in liver enzymes only in allopurinol group
Farajzadeh *et al.*, 2015 (26)	20.74 16.3	*L. tropica*	IM meglumine antimoniate + cryotherapy (40) Oral terbinafine + cryotherapy (40)	52.5% 37.5%	12	No major AEs were observed in either group
Firooz *et al.*, 2006 (27)	27	*L. tropica*	IM meglumine antimoniate + imiquimod cream (59) IM meglumine antimoniate (60)	44.1% 40%	20	Three patients experienced pruritus and burning sensation No AEs were reported
Hanif *et al.*, 2016 (28)	39	Not specified	Oral chloroquine (38)	100%	16	No systemic AEs
Hepburn *et al.*, 1994 (29)	23.8 23.5	*L. mexicana* and *L. braziliensis*	IV paromomycin (17) IV sodium stibogluconate (17)	59% 88%	24	Well tolerated with mild AEs such as muscle aches and a papular rash in one patient More significant AEs, including muscle aches, nausea, and bone marrow suppression
Jaffar *et al.*, 2006 (30)	3-65	Not specified	Oral rifampicin (34)	61.8%	12	One patient had transient elevated liver enzymes
Khattak *et al.*, 2021 (31)	27.4	*L. tropica*	Oral allopurinol + IV meglumine antimoniate (56) IV meglumine antimoniate (56)	83.9% 73.2%	8	No major AEs were reported in either group
Kochar *et al.*, 2000 (32)	29.5	Not specified	Oral rifampicin (23)	73.9%	4	No major AEs
Kochar *et al.*, 2006 (33)	16-70	*L. tropica*	Oral rifampicin + omeprazole (23)	69.6%	6	No major AEs
Krolewiecki *et al.*, 2007 (34)	33 37	*L. braziliensis*	Oral azithromycin (22) IV or IM meglumine antimoniate (23)	45.5% 82.6%	52	Well-tolerated with mild gastrointestinal AEs 78% experienced musculoskeletal AEs
Lago *et al.*, 2024 (35)	18-50	*L. braziliensis*	IV meglumine antimoniate + topical rSm29 (31) IV meglumine antimoniate (30)	71% 43%	12	56% experienced joint and muscle pain, with some cases of mild itching and small papular lesions that self-healed 54% experienced joint and muscle pain

Layegh *et al.*, 2007 (36)	21.77	*L. tropica* and *L. major*	Oral azithromycin (20, 29 lesions)	10.3%	16	Two patients experienced gastrointestinal symptoms (nausea, vomiting), leading to discontinuation
	22.78		IM meglumine antimoniate (27, 58 lesions)	34.5%		Three patients reported myalgia and erythema at the injection site
Lobo *et al.*, 2006 (37)	36	*L. braziliensis*	IV meglumine antimoniate (20)	10%	4	Mild secondary bacterial infections in one patient
López *et al.*, 2012 (38)	Not specified	*L. braziliensis* and *L. panamensis*	IM meglumine antimoniate (121)	85%	24	Myalgia, arthralgia, headache, and fever
López *et al.*, 2022 (39)	34.3	*L. braziliensis* and *L. panamensis*	Oral miltefosine + thermotherapy (62)	83.9%	24	Vomiting (31.8%) and liver enzyme elevation (28.8%)
Lyra *et al.*, 2023 (40)	≥ 13	Not specified	IV or IM meglumine antimoniate (59)	67.8%	24	80.6% clinical AEs, 73.1% laboratory AEs, and 25.4% electrocardiographic AEs. Ten participants were discontinued due to severe or persistent AEs
Machado *et al.*, 2010 (41)	22	*L. braziliensis*	Oral miltefosine (60)	75%	24	Vomiting, nausea, and abdominal pain with 78.3% rate of AEs
	23		IV meglumine antimoniate (30)	53.3%		Arthralgia, myalgia, and fever with 76.7% rate of AEs
Machado *et al.*, 2018 (42)	19-53	*L. braziliensis*	IV meglumine antimoniate (15)	40%	24	Mild AEs were observed in all groups, with arthralgia, myalgia, and fever being the most common. One patient in the oral tamoxifen group dropped out due to headache and palpitations.
			IV meglumine antimoniate + topical tamoxifen (11)	36.4%		
			IV meglumine antimoniate + topical tamoxifen (12)	58%		
Machado *et al.*, 2021 (43)	33	*L. braziliensis*	IV meglumine antimoniate (45)	44.4%	24	65% rate of AEs, such as arthralgia and/or myalgia, fever, and headache
	36		Oral miltefosine (47)	76.6%		76% rate of AEs, such as nausea, vomiting, and headache
	33		Oral miltefosine + topical GM-CSF (41)	75.6%		79% rate of AEs, such as nausea, vomiting, and headache
Martinez *et al.*, 1992 (44)	11-40	*L. panamensis*	IV meglumine antimoniate (33)	36%	52	No major AEs
			Oral allopurinol (25)	80%		
			Oral allopurinol + IV meglumine antimoniate (35)	74%		

Martinez *et al.*, 1997 (45)	18-57	*L. braziliensis* and *L. panamensis*	IV sodium stibogluconate + oral allopurinol (51) IV sodium stibogluconate (49)	71% 39%	52	More cases of eosinophilia and rash, but no significant toxic effects One patient stopped treatment due to severe chemical hepatitis with neurological manifestations
Mendes *et al.*, 2021 (46)	18-65	*L. guyanensis*	Oral miltefosine (50) Oral miltefosine + topical GM-CSF (50) IV meglumine antimoniate (50)	66% 58% 50%	24	Common AEs were nausea (85%) and vomiting (39%) Frequent arthralgia (51%) and myalgia (48%). One severe AE led to treatment discontinuation
Miranda-Verástegui *et al.*, 2005 (47)	14.1 19.25	*L. peruviana* and *L. braziliensis*	IM or IV meglumine antimoniate + imiquimod cream (18) IM or IV meglumine antimoniate (20)	72% 75%	52	Neurotoxic effects of antimony therapy have been reported Mild erythema was more common, but no severe AEs were reported
Miranda-Verástegui *et al.*, 2009 (48)	25 25.9	*L. braziliensis*, *L. peruviana*, and *L. guyanensis*	IV sodium stibogluconate + imiquimod cream (40) IV sodium stibogluconate (40)	75% 58%	52	Similar mild AEs were noted in both groups.
Mohebali *et al.*, 2007 (49)	20.2 16.8	*L. major*	Oral miltefosine (28) IM meglumine antimoniate (30)	92.9% 83.3%	24	Nausea and vomiting were observed in miltefosine group. No significant difference in other AEs reporting was seen between the two groups.
Momeni *et al.*, 2002 (50)	20.2 23.6	*L. major*	IV or IM meglumine antimoniate (35) IV or IM meglumine antimoniate (low-dose) + oral allopurinol (31)	72.4% 80.6%	4	Skin eruption in one patient and generalized muscle pain and weakness in four patients Mild abdominal pain and nausea in six patients
Munir *et al.*, 2008 (51)	14-55	*L. tropica*	IM meglumine antimoniate (20) IM + IL meglumine antimoniate (20)	55% 75%	12	Combination group experienced more AEs, including myalgia, arthralgia, headache, urticaria, and one case of cardiomyopathy, which resolved after treatment discontinuation
Nassiri-Kashani *et al.*, 2005 (52)	12-60	*L. major*	Oral itraconazole (70)	95.7%	12	Gastrointestinal complaints and headaches

Navin *et al.*, 1992 (53)	20	*L. braziliensis* and *L. mexicana*	IV sodium stibogluconate (32)	90.6%	52	Elevated liver enzymes and electrocardiographic changes, but these normalized by week 9
			Oral ketoconazole (32)	56.3%		Gastrointestinal complaints
Neves *et al.*, 2011 (54)	29.3	*L. guyanensis* and *L. braziliensis*	IM pentamidine (58)	62.1%	24	35.1% experienced injection site pain
	28.5		IV or IM meglumine antimoniate (58)	60.3%		20.3% experienced arthralgia, 9.5% had headaches
Ozgoztasi *et al.*, 1997 (55)	≤ 10	Not specified	Oral ketoconazole (32)	0%	4	No AEs
Parhizkar *et al.*, 2024 (56)	29.8	*L. major*	Oral fluconazole + cryotherapy (28, 49 lesions)	53.1%	4	No AEs
Pennini *et al.*, 2023 (57)	37.9	*L. guyanensis*	IM pentamidine (24)	100%	24	Mild AEs related to pentamidine were reported, such as bitter taste, dyspepsia, abdominal pain, and mild hypoglycemia. There were no AEs related to tamoxifen.
	36.6		IM pentamidine + oral tamoxifen (25)	72%		
Prates *et al.*, 2017 (58)	27.9	*L. braziliensis*	Oral fluconazole (27)	22.2%	24	Mild AEs with similar frequencies in both groups (fluconazole: 37%, sodium stibogluconate: 34.6%). One patient in the fluconazole group discontinued due to malaise, headache, and dizziness.
	35		IV sodium stibogluconate (26)	53.8%		
Rahman *et al.*, 2023 (59)	28.82	Not specified	IM meglumine antimoniate (33)	33.3%	4	Injection site pain (39.3%), abscess formation (21.2%), and ECG changes (24.2%)
	32.09		Oral miltefosine (33)	12.1%		Nausea and vomiting (63.6%), abdominal fullness (33.3%), and fever (15.2%)
Ribeiro *et al.*, 2021 (60)	18-65	*L. braziliensis*	IV meglumine antimoniate (14)	43%	24	2 patients with discontinuation of treatment protocol due to AEs: one from the oral tamoxifen group and the other from the topical tamoxifen group
			IV meglumine antimoniate + oral tamoxifen (10)	70%		
			IV meglumine antimoniate + topical tamoxifen (10)	40%		
Rubiano *et al*, 2012 (61)	2-12	*L. panamensis* and *L. guyanensis*	Oral miltefosine (55)	87.3%	26	Nausea, vomiting, and abdominal pain
			IM meglumine antimoniate (56)	71.4%		Elevated liver enzymes, fever, and loss of appetite

Sadeghian *et al.*, 2006 (62)	27	*L. major*	IM meglumine antimoniate + oral pentoxifylline (32)	81.3%	12	No significant AEs
	31		IM meglumine antimoniate (31)	51.6%		A pruritic rash in one patient
Saenz *et al.*, 1990 (63)	25	*L. panamensis*	Oral ketoconazole (22)	73%	52	Mild, reversible hepatocellular enzyme elevation in 27% of patients, and a reversible decrease in serum testosterone levels
	34		IM sodium stibogluconate (19)	68%		47% had mild hepatocellular enzyme elevation, and most patients experienced pain at the injection site, myalgia, or headache
Salmanpour *et al.*, 2001 (64)	20.7	*L. tropica* and *L. major*	Oral ketoconazole (64)	89%	24	Mild nausea and abdominal pain in some patients; two cases of increased liver enzymes (resolved post-treatment)
Sattar *et al.*, 2020 (65)	26.76	Not specified	Oral zinc sulphate (34)	58.8%	6	Mild nausea and abdominal pain in some patients
Shahzad *et al.*, 2020 (66)	34.26	Not specified	Oral azithromycin (28)	67.9%	12	No significant AEs
Shanehsaz *et al.*, 2015 (67)	25.2	Not specified	IM meglumine antimoniate (30, 90 lesions)	91.1%	12	Higher incidence of AEs, including musculoskeletal pain, nausea, and minor skin reactions
			IM meglumine antimoniate (low-dose) + oral cimetidine (30, 150 lesions)	84.7%		Slightly higher AEs rate compared to low-dose meglumine antimoniate alone
			IM meglumine antimoniate (low-dose) (30, 120 lesions)	78.3%		Lower incidence of AEs
Soto *et al.*, 2004 (68)	24.9	*L. panamensis* and *L. braziliensis* and *L. mexicana*	Oral miltefosine (82)	73.2%	24	Nausea (36%) and vomiting (31%), with mild increases in creatinine observed in 33% of participants
Soto-Mancipe *et al.*, 1993 (69)	18-25	*L. panamensis* and *L. braziliensis* and *L. mexicana*	IM meglumine antimoniate (23)	91%	52	Nine patients (39%) reported subjective. AEs, including arthralgia and myalgia.
			IM pentamidine (24)	96%		Eight patients (30%) reported mild to moderate subjective AEs, including myalgias, nausea and headache.
			Oral itraconazole (20)	5%		Not reported.

Toledo Junior *et al.*, 2014 (70)	14-65	*L. braziliensis*	IM or IV meglumine antimoniate (18) Oral azithromycin (21)	72.2% 23.8%	12	Higher AE incidence (62.5%), including myalgia and arthralgia Fewer AEs, mainly gastrointestinal issues
Vélez *et al.*, 2010 (71)	19-38	*L. panamensis* and *L. braziliensis*	Oral miltefosine (122) IM meglumine antimoniate (121)	69.8% 85.1%	24	Gastrointestinal symptoms such as nausea, vomiting, and abdominal pain Musculoskeletal pain, fever, and higher organ toxicity
Yazdanpanah *et al.*, 2011 (72)	32 25	*L. tropica*	Oral zinc sulfate (26, 43 lesions) IM meglumine antimoniate (74, 127 lesions)	30.2% 35.5%	6	No significant AEs Serious AEs, such as severe muscle pain and injection site reactions, leading to discontinuation for some patients
Zabolinejad *et al.*, 2022 (73)	31.8 35.8	Not specified	Oral clarithromycin (10, 49 lesions) IM or IV meglumine antimoniate (10, 45 lesions)	100% 96.3%	52	No AEs reported in either group
Zerehsaz *et al.*, 1999 (74)	1.5-69	Not specified	IM meglumine antimoniate (85)	27.1%	6	Cutaneous AEs, including urticaria and generalized pruritus

AEs, Adverse events; IM, intramuscular; IL, intralesional; IV, intravenous; GM-CSF, granulocyte-macrophage colony-stimulating factor; ECG, electrocardiogram.

Table S2. Risk of Bias assessment of randomized clinical trials. The risk-of-bias tool version 2 (RoB2), which is the recommended tool to assess the risk of bias in randomized trials included in Cochrane Reviews, was applied for the evaluation. D1, Randomization process; D2, deviations from the intended interventions; D3, missing outcome data; D4, measurement of the outcome; D5, selection of the reported result; Green, low risk; Yellow, some concerns; Red, high risk.

Study	Risk of bias					
	D1	D2	D3	D4	D5	Overall
Abtahi-Naeini *et al.*, 2021 (1)	!	+	+	+	!	!
Adam *et al.*, 2009 (2)	+	+	+	+	!	!
Ahmed *et al.*, 2022 (3)	+	+	+	+	!	!
Alfahaad, 2020 (4)	!	+	-	!	!	-
Al-Fouzan *et al.*, 1991 (5)	!	+	+	+	!	!
Alkhawajah *et al.*, 1997 (6)	!	!	-	+	!	-
Almeida *et al.*, 1999 (7)	!	+	+	+	!	!
Alrajhi *et al.*, 2002 (8)	+	+	+	+	!	!
Arana *et al.*, 1994 (9)	!	+	+	+	!	!
Arevalo *et al.*, 2007 (10)	!	+	+	!	!	!
Armijos *et al.*, 2004 (11)	+	+	!	+	!	!
Aronson *et al.*, 2010 (12)	+	+	+	+	+	+
Bashir *et al.*, 2019 (13)	+	+	!	+	+	!
Brito *et al.*, 2014 (14)	!	+	+	+	!	!
Brito *et al.*, 2017 (15)	!	+	+	+	!	!
Castro *et al.*, 2022 (16)	+	+	+	+	+	+
Celes *et al.*, 2022 (17)	+	+	+	+	+	+
Chrusciak-Talhari *et al.*, 2011 (18)	+	+	+	+	+	+
D'Oliveira Jr. *et al.*, 1997 (19)	!	+	-	+	!	-
Dastgheib *et al.*, 2012 (20)	!	+	+	-	!	-
Dogra *et al.*, 1990 (21)	!	+	+	-	!	-
Dogra *et al.*, 1994 (22)	+	+	+	+	!	!
Dogra *et al.*, 1996 (23)	!	-	+	!	!	-
El-Sayed *et al.*, 2010 (24)	!	+	+	!	+	!
Esfandiarpour *et al.*, 2002 (25)	!	+	+	!	!	!
Farajzadeh *et al.*, 2015 (26)	+	+	+	+	+	+
Firooz *et al.*, 2006 (27)	+	+	+	+	+	+
Hanif *et al.*, 2016 (28)	+	+	+	!	+	!
Hepburn *et al.*, 1994 (29)	!	+	+	!	+	!
Jaffar *et al.*, 2006 (30)	!	+	+	+	+	!
Khattak *et al.*, 2021 (31)	!	+	+	!	+	!
Kochar *et al.*, 2000 (32)	+	+	+	+	+	+
Kochar *et al.*, 2006 (33)	!	+	+	+	+	!
Krolewiecki *et al.*, 2007 (34)	+	+	+	!	!	!
Lago *et al.*, 2024 (35)	+	+	+	!	+	!
Layegh *et al.*, 2007 (36)	!	+	+	!	+	!
Lobo *et al.*, 2006 (37)	!	+	+	!	!	!
López *et al.*, 2012 (38)	+	+	+	!	+	!
López *et al.*, 2022 (39)	+	+	+	+	+	+
Lyra *et al.*, 2023 (40)	+	+	+	!	+	!
Machado *et al.*, 2010 (41)	+	+	+	+	+	+
Machado *et al.*, 2018 (42)	+	+	+	+	+	+
Machado *et al.*, 2021 (43)	+	+	+	+	+	+
Martinez *et al.*, 1992 (44)	+	+	+	!	+	!
Martinez *et al.*, 1997 (45)	+	+	+	!	+	!
Mendes *et al.*, 2021 (46)	+	+	+	!	+	!

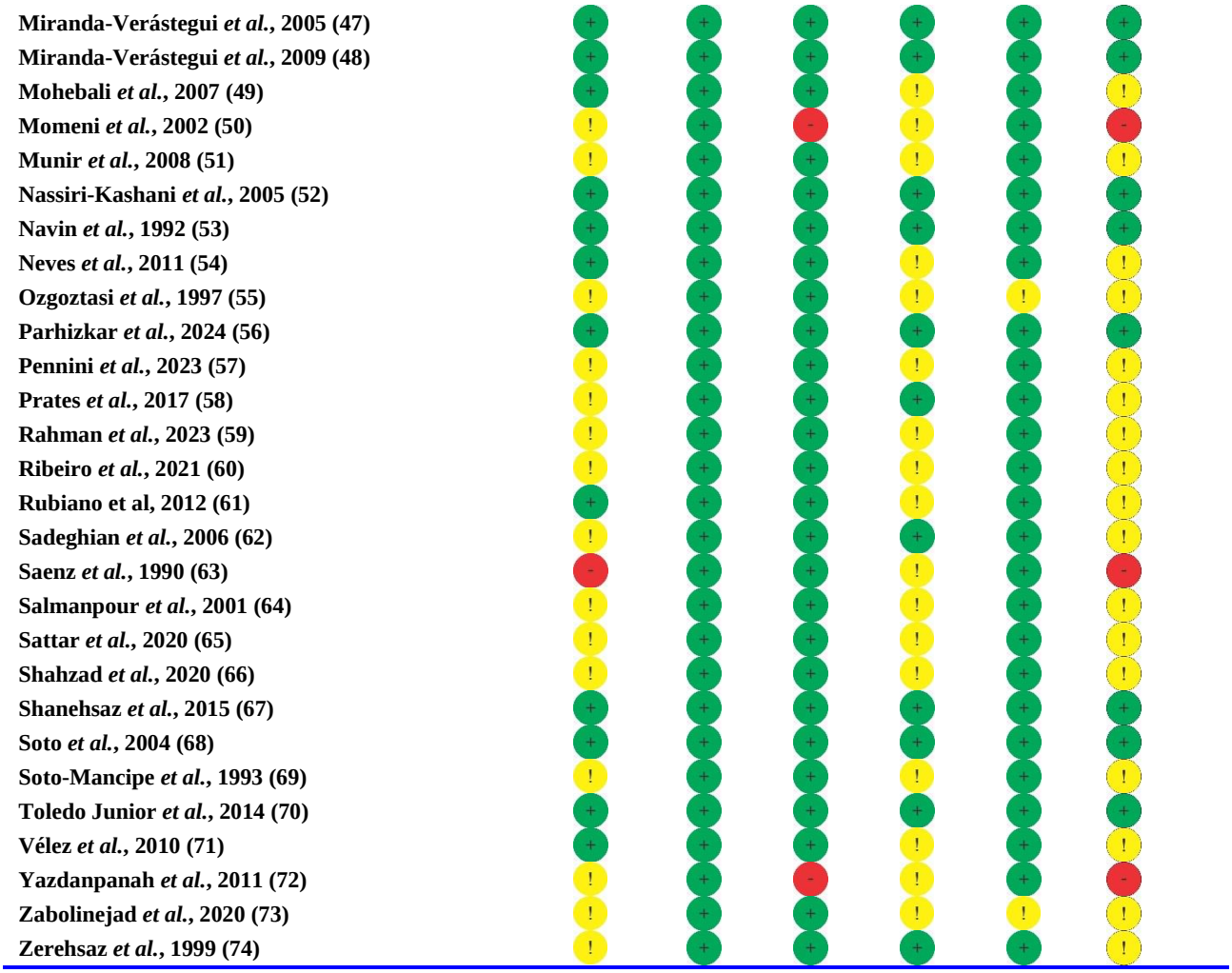

Miranda-Verástegui et al., 2005 (47)
Miranda-Verástegui et al., 2009 (48)
Mohebali et al., 2007 (49)
Momeni et al., 2002 (50)
Munir et al., 2008 (51)
Nassiri-Kashani et al., 2005 (52)
Navin et al., 1992 (53)
Neves et al., 2011 (54)
Ozgoztasi et al., 1997 (55)
Parhizkar et al., 2024 (56)
Pennini et al., 2023 (57)
Prates et al., 2017 (58)
Rahman et al., 2023 (59)
Ribeiro et al., 2021 (60)
Rubiano et al, 2012 (61)
Sadeghian et al., 2006 (62)
Saenz et al., 1990 (63)
Salmanpour et al., 2001 (64)
Sattar et al., 2020 (65)
Shahzad et al., 2020 (66)
Shanehsaz et al., 2015 (67)
Soto et al., 2004 (68)
Soto-Mancipe et al., 1993 (69)
Toledo Junior et al., 2014 (70)
Vélez et al., 2010 (71)
Yazdanpanah et al., 2011 (72)
Zabolinejad et al., 2020 (73)
Zerehsaz et al., 1999 (74)

Table S3. Node-splitting analysis for local inconsistency under the random-effects model for the sensitivity NMA (excluding high risk studies)

Treatment comparison	k	Direct evidence proportion	NMA RR	Direct RR	Indirect RR	RoR	z	*P*-value
Systemic allopurinol versus placebo/control	1	0.07	4.77	28.38	4.22	6.73	1.29	0.197
Systemic azithromycin versus placebo/control	1	0.53	1.61	1.58	1.63	0.97	-0.05	0.958
Systemic fluconazole versus placebo/control	1	0.76	1.87	2.33	0.94	2.48	1.26	0.207
Systemic ketoconazole versus placebo/control	2	0.82	1.9	1.88	1.98	0.95	-0.06	0.954
Systemic meglumine antimoniate versus placebo/control	3	0.24	3.56	6.95	2.88	2.41	1.55	0.121
Systemic meglumine antimoniate + systemic allopurinol versus	1	0.05	5.61	26.5	5.17	5.13	1.11	0.265
Systemic miltefosine versus placebo/control	2	0.55	3.88	2.45	6.72	0.36	-2.09	**0.037**
Systemic pentamidine versus placebo/control	1	0.19	3.89	7.67	3.32	2.31	1.04	0.3
Systemic sodium stibogluconate versus placebo/control	2	0.63	2.92	1.87	6.23	0.3	-1.8	0.072
Systemic allopurinol versus systemic meglumine antimoniate	2	0.82	1.34	1.45	0.94	1.55	0.61	0.54
Systemic allopurinol versus systemic meglumine antimoniate + systemic allopurinol	2	0.92	0.85	0.76	3.45	0.22	-1.59	0.112
Systemic azithromycin versus systemic meglumine antimoniate	2	0.7	0.45	0.46	0.44	1.03	0.05	0.958
Systemic fluconazole versus systemic sodium stibogluconate	1	0.51	0.64	0.41	1.02	0.4	-1.26	0.207
Systemic itraconazole versus systemic meglumine antimoniate	1	0.12	0.33	0.06	0.41	0.16	-1.71	0.088
Systemic itraconazole versus systemic pentamidine	1	0.16	0.3	0.06	0.41	0.15	-1.7	0.088
Systemic ketoconazole versus systemic sodium stibogluconate	2	0.94	0.65	0.74	0.09	7.82	1.8	0.071
Systemic meglumine antimoniate versus systemic meglumine antimoniate + systemic allopurinol	3	0.99	0.63	0.65	0.09	7.48	0.88	0.378
Systemic meglumine antimoniate versus systemic miltefosine	8	0.93	0.92	0.86	2.36	0.36	-2.09	**0.037**
Systemic meglumine antimoniate versus systemic pentamidine	2	0.97	0.92	0.99	0.08	11.68	1.9	0.057

NMA, Network meta-analysis; RR, relative risk; RoR, ratio of ratios; k, number of studies included in the comparison; z, z-statistic; *P*-value, probability value.

Table S4. Node-splitting analysis for local inconsistency under the random-effects model for the overall NMA

Treatment comparison	k	Direct evidence proportion	NMA RR	Direct RR	Indirect RR	RoR	z	*P*-value
Systemic allopurinol versus placebo/control	1	0.06	4.08	28.38	3.58	7.93	1.4	0.162
Systemic azithromycin versus placebo/control	1	0.53	1.63	1.58	1.67	0.95	-0.1	0.923
Systemic fluconazole versus placebo/control	1	0.75	1.87	2.33	0.97	2.39	1.21	0.226
Systemic ketoconazole versus placebo/control	3	0.79	2.29	2.18	2.73	0.8	-0.29	0.772
Systemic meglumine antimoniate versus placebo/control	3	0.24	3.63	6.95	2.95	2.35	1.5	0.134
Systemic meglumine antimoniate + systemic allopurinol versus placebo/control	1	0.04	5	26.5	4.62	5.73	1.19	0.233
Systemic miltefosine versus placebo/control	2	0.54	3.94	2.45	6.95	0.35	-2.13	**0.033**
Systemic pentamidine versus placebo/control	1	0.19	4	7.67	3.42	2.24	0.99	0.322
Systemic sodium stibogluconate versus placebo/control	3	0.63	2.94	2.23	4.68	0.48	-1.15	0.249
Systemic allopurinol versus systemic meglumine antimoniate	3	0.78	1.12	1.25	0.77	1.63	0.76	0.45
Systemic allopurinol versus systemic meglumine antimoniate + systemic allopurinol	2	0.88	0.82	0.75	1.46	0.51	-0.85	0.393
Systemic azithromycin versus systemic meglumine antimoniate	2	0.7	0.45	0.45	0.43	1.06	0.1	0.923
Systemic fluconazole versus systemic sodium stibogluconate	1	0.5	0.64	0.41	0.99	0.42	-1.21	0.226
Systemic itraconazole versus systemic meglumine antimoniate	1	0.12	0.39	0.06	0.5	0.13	-1.89	0.059
Systemic itraconazole versus systemic pentamidine	1	0.16	0.36	0.06	0.5	0.12	-1.89	0.059
Systemic ketoconazole versus systemic sodium stibogluconate	3	0.96	0.78	0.84	0.1	8.28	1.83	0.067
Systemic meglumine antimoniate versus systemic meglumine antimoniate + systemic allopurinol	4	0.99	0.73	0.72	2.77	0.26	-0.82	0.415
Systemic meglumine antimoniate versus systemic miltefosine	8	0.93	0.92	0.86	2.44	0.35	-2.13	**0.033**
Systemic meglumine antimoniate versus systemic pentamidine	2	0.97	0.91	0.99	0.08	13.15	1.98	**0.048**

NMA, Network meta-analysis; RR, relative risk; RoR, ratio of ratios; k, number of studies included in the comparison; z, z-statistic; *P*-value, probability value.

REFERENCES

1. Abtahi-Naeini B, Hadian S, Sokhanvari F, Hariri A, Varshosaz J, Shahmoradi Z, *et al*. Effect of adjunctive topical liposomal azithromycin on systemic azithromycin on Old World cutaneous leishmaniasis: a pilot clinical study. Iran J Pharm Res. 2021;20(2):383-389.
DOI: 10.22037/ijpr.2020.113710.14445.
2. Adam I, Hagelnur AA. Artesunate plus sulfamethoxypyrazine/pyrimethamine for the treatment of cutaneous leishmaniasis: a double-blind, placebo-controlled clinical trial. Int J Antimicrob Agents. 2009;34(4):380-381.
DOI: 10.1016/j.ijantimicag.2009.03.013.
3. Ahmed N, Malik S, Tahir M, Rahman A, Fayyaz I, Raza N, *et al*. Comparison of oral dapsone with intramuscular meglumine antimoniate in cutaneous leishmaniasis. J Ayub Med Coll Abbottabad. 2022;34(4):802-806.
DOI: 10.55519/JAMC-04-10265.
4. Alfahaad HA. Combined intralesional and intramuscular sodium stibogluconate appears more effective in the treatment of localized cutaneous leishmaniasis lesions: an experimental study. J Pak Assoc Dermatol. 2020;30(3):396-401.
DOI: 10.66344/jpad.v30i3.1450.
5. al-Fouzan AS, al Saleh QA, Najem NM, Rostom AI. Cutaneous leishmaniasis in Kuwait: clinical experience with itraconazole. Int J Dermatol. 1991;30(7):519-521.
DOI: 10.1111/j.1365-4362.1991.tb04878.x.
6. Alkhawajah AM, Larbi E, al-Gindan Y, Abahussein A, Jain S. Treatment of cutaneous leishmaniasis with antimony: intramuscular versus intralesional administration. Ann Trop Med Parasitol. 1997;91(8):899-905.
DOI: 10.1080/00034989760284.
7. Almeida R, D'Oliveira A Jr, Machado P, Bacellar O, Ko AI, de Jesus AR, *et al*. Randomized, double-blind study of stibogluconate plus human granulocyte macrophage colony-stimulating factor versus stibogluconate alone in the treatment of cutaneous leishmaniasis. J Infect Dis. 1999;180(5):1735-1737.
DOI: 10.1086/315082.
8. Alrajhi AA, Ibrahim EA, De Vol EB, Khairat M, Faris RM, Maguire JH. Fluconazole for the treatment of cutaneous leishmaniasis caused by *Leishmania major*. N Engl J Med. 2002;346(12):891-895.
DOI: 10.1056/NEJMoa011882.
9. Arana BA, Navin TR, Arana FE, Berman JD, Rosenkaimer F. Efficacy of a short course (10 days) of high-dose meglumine antimonate with or without interferon-gamma in treating cutaneous leishmaniasis in Guatemala. Clin Infect Dis. 1994;18(3):381-384.
DOI: 10.1093/clinids/18.3.381.
10. Arevalo I, Tulliano G, Quispe A, Spaeth G, Matlashewski G, Llanos-Cuentas A, *et al*. Role of imiquimod and parenteral meglumine antimoniate in the initial treatment of cutaneous leishmaniasis. Clin Infect Dis. 2007;44(12):1549-1554.
DOI: 10.1086/518172.
11. Armijos RX, Weigel MM, Calvopina M, Mancheno M, Rodriguez R. Comparison of the effectiveness of two topical paromomycin treatments versus meglumine antimoniate for New World cutaneous leishmaniasis. Acta Trop. 2004;91(2):153-160.
DOI: 10.1016/j.actatropica.2004.03.009.
12. Aronson NE, Wortmann GW, Byrne WR, Howard RS, Bernstein WB, Marovich MA, *et al*. A randomized controlled trial of local heat therapy versus intravenous sodium stibogluconate for the treatment of cutaneous *Leishmania major* infection. PLoS Negl Trop Dis. 2010;4(3):e628.
DOI: 10.1371/journal.pntd.0000628.
13. Bashir U, Tahir M, Anwar MI, Manzoor F. Comparison of intralesional meglumine antimonite along with oral itraconazole to intralesional meglumine antimonite in the treatment of cutaneous leishmaniasis. Pak J Med Sci. 2019;35(6):1669-1673.
DOI: 10.12669/pjms.35.6.363.
14. Brito G, Dourado M, Polari L, Celestino D, Carvalho LP, Queiroz A, *et al*. Clinical and immunological outcome in cutaneous leishmaniasis patients treated with pentoxifylline. Am J Trop Med Hyg. 2014;90(4):617-620.
DOI: 10.4269/ajtmh.12-0729.
15. Brito G, Dourado M, Guimaraes Guimarães LH, Meireles E, Schriefer A, de Carvalho EM, *et al*. Oral pentoxifylline associated with pentavalent antimony: a randomized trial for cutaneous leishmaniasis. Am J Trop Med Hyg. 2017;96(5):1155-1159.
DOI: 10.4269/ajtmh.16-0435.
16. Castro MDM, Cossio A, Navas A, Fernandez O, Valderrama L, Cuervo-Pardo L, *et al*. Pentoxifylline in the treatment of cutaneous leishmaniasis: a randomized clinical trial in Colombia. Pathogens. 2022;11(3):378.
DOI: 10.3390/pathogens11030378.
17. Celes FS, Barud HS, Viana SM, Borba PB, Machado PRL, Carvalho EM, *et al*. A pilot and open trial to evaluate topical bacterial cellulose bio-curatives in the treatment of cutaneous leishmaniasis caused by L. braziliensis. Acta Trop. 2022;225:106192.

DOI: 10.1016/j.actatropica.2021.106192.
18. Chrusciak-Talhari A, Dietze R, Chrusciak Talhari C, da Silva RM, Gadelha Yamashita EP, de Oliveira Penna G, *et al*. Randomized controlled clinical trial to assess efficacy and safety of miltefosine in the treatment of cutaneous leishmaniasis caused by *Leishmania* (Viannia) *guyanensis* in Manaus, Brazil. Am J Trop Med Hyg. 2011;84(2):255-260.
DOI: 10.4269/ajtmh.2011.10-0155.
19. D'Oliveira Junior A, Machado PR, Carvalho EM. Evaluating the efficacy of allopurinol for the treatment of cutaneous leishmaniasis. Int J Dermatol. 1997;36(12):938-940.
DOI: 10.1046/j.1365-4362.1997.00308.x.
20. Dastgheib L, Naseri M, Mirashe Z. Both combined oral azithromycin plus allopurinol and intramuscular glucantime yield low efficacy in the treatment of Old World cutaneous leishmaniasis: a randomized controlled clinical trial. Int J Dermatol. 2012;51(12):1508-1511.
DOI: 10.1111/j.1365-4632.2012.05610.x.
21. Dogra J, Aneja N, Lal BB, Mishra SN. Cutaneous leishmaniasis in India: clinical experience with itraconazole (R51 211 Janssen). Int J Dermatol. 1990;29(9):661-662.
DOI: 10.1111/j.1365-4362.1990.tb02593.x.
22. Dogra J, Aneja N. Leishmaniasis and itraconazole: a controlled clinical trial on cutaneous subtypes. Int J Antimicrob Agents. 1994;4(4):309-311.
DOI: 10.1016/0924-8579(94)90031-0.
23. Dogra J, Saxena VN. Itraconazole and leishmaniasis: a randomized double-blind trial in cutaneous disease. Int J Parasitol. 1996;26(12):1413-1415.
DOI: 10.1016/S0020-7519(96)00128-2.
24. El-Sayed M, Anwar AE. Intralesional sodium stibogluconate alone or its combination with either intramuscular sodium stibogluconate or oral ketoconazole in the treatment of localized cutaneous leishmaniasis: a comparative study. J Eur Acad Dermatol Venereol. 2010;24(3):335-340.
DOI: 10.1111/j.1468-3083.2009.03417.x.
25. Esfandiarpour I, Alavi A. Evaluating the efficacy of allopurinol and meglumine antimoniate (Glucantime) in the treatment of cutaneous leishmaniasis. Int J Dermatol. 2002;41(8):521-524.
DOI: 10.1046/j.1365-4362.2002.01526.x.
26. Farajzadeh S, Esfandiarpour I, Haghdoost AA, Mohammadi S, Mohebbi A, Mohebbi E, *et al*. Comparison between combination therapy of oral terbinafine and cryotherapy versus systemic meglumine antimoniate and cryotherapy in cutaneous leishmaniasis: a randomized clinical trial. Iran J Parasitol. 2015;10(1):1-8.
PMID: 25904940.
27. Firooz A, Khamesipour A, Ghoorchi MH, Nassiri-Kashani M, Eskandari SE, Khatami A, *et al*. Imiquimod in combination with meglumine antimoniate for cutaneous leishmaniasis: a randomized assessor-blind controlled trial. Arch Dermatol. 2006;142(12):1575-1579.
DOI: 10.1001/archderm.142.12.1575.
28. Hanif MM, Akram K, Mustafa G. Intralesional versus oral chloroquine in cutaneous leishmaniasis: comparison of outcome, duration of treatment and total dose of drug. J Coll Physicians Surg Pak. 2016;26(4):260-262.
PMID: 27097693.
29. Hepburn NC, Tidman MJ, Hunter JA. Aminosidine (paromomycin) versus sodium stibogluconate for the treatment of American cutaneous leishmaniasis. Trans R Soc Trop Med Hyg. 1994;88(6):700-703.
DOI: 10.1016/0035-9203(94)90237-2.
30. Jaffar H. Rifampicin in cutaneous leishmaniasis-a therapeutic trial in Saudi Arabia. J Pak Assoc Dermatol. 2006;16(1):4-9.
DOI: 10.66344/jpad.16.1.2006.838.
31. Khattak JI, Ullah G, Khan QU. Efficacy of combined parenteral meglumine antimoniate and oral allopurinol with meglumine antimoniate alone in treatment of cutaneous leishmaniasis. J Pak Assoc Dermatol. 2021;30(4):644-649.
DOI: 10.66344/jpad.v30i4.1483.
32. Kochar DK, Aseri S, Sharma BV, Bumb RA, Mehta RD, Purohit SK. The role of rifampicin in the management of cutaneous leishmaniasis. QJM. 2000;93(11):733-737.
DOI: 10.1093/qjmed/93.11.733.
33. Kochar DK, Saini G, Kochar SK, Sirohi P, Bumb RA, Mehta RD, *et al*. A double blind, randomized placebo controlled trial of rifampicin with omeprazole in the treatment of human cutaneous leishmaniasis. J Vector Borne Dis. 2006;43(4):161-167.
PMID: 17175700.
34. Krolewiecki AJ, Romero HcD, Cajal SP, Abraham D, Mimori T, Matsumoto T, *et al*. A randomized clinical trial comparing oral azithromycin and meglumine antimoniate for the treatment of American cutaneous leishmaniasis caused by *Leishmania* (Viannia) *braziliensis*. Am J Trop Med Hyg. 2007;77(4):640-646.
DOI: 10.4269/ajtmh.2007.77.640.

35. Lago T, Peixoto F, Mambelli F, Carvalho LP, Guimaraes Guimarães LH, Carvalho AM, *et al*. Use of topical rSm29 in combination with intravenous meglumine antimoniate in the treatment of cutaneous leishmaniasis: a randomized controlled trial. Int J Infect Dis. 2024;147:107206.
DOI: 10.1016/j.ijid.2024.107206.
36. Layegh P, Yazdanpanah MJ, Vosugh EM, Pezeshkpoor F, Shakeri MT, Moghiman T. Efficacy of azithromycin versus systemic meglumine antimoniate (Glucantime) in the treatment of cutaneous leishmaniasis. Am J Trop Med Hyg. 2007;77(1):99-101.
DOI: 10.4269/ajtmh.2007.77.99.
37. Lobo IM, Soares MB, Correia TM, de Freitas LA, Oliveira MI, Nakatani M, *et al*. Heat therapy for cutaneous leishmaniasis elicits a systemic cytokine response similar to that of antimonial (Glucantime) therapy. Trans R Soc Trop Med Hyg. 2006;100(7):642-649.
DOI: 10.1016/j.trstmh.2005.08.011.
38. López L, Robayo M, Vargas M, Velez Vélez ID. Thermotherapy: an alternative for the treatment of American cutaneous leishmaniasis. Trials. 2012;13:58.
DOI: 10.1186/1745-6215-13-58.
39. López L, Valencia B, Alvarez F, Ramos AP, Llanos-Cuentas A, Echevarria J, *et al*. A phase II multicenter randomized study to evaluate the safety and efficacy of combining thermotherapy and a short course of miltefosine for the treatment of uncomplicated cutaneous leishmaniasis in the New World. PLoS Negl Trop Dis. 2022;16(3):e0010238.
DOI: 10.1371/journal.pntd.0010238.
40. Lyra MR, Oliveira LFA, Schubach AO, Sampaio RNR, Rodrigues BC, Hueb M, *et al*. A randomized, controlled, noninferiority, multicenter trial of systemic versus intralesional treatment with meglumine antimoniate for cutaneous leishmaniasis in Brazil. Clin Infect Dis. 2023;77(4):574-582.
DOI: 10.1093/cid/ciad253.
41. Machado PR, Ampuero J, Guimaraes Guimarães LH, Villasboas L, Rocha AT, Schriefer A, *et al*. Miltefosine in the treatment of cutaneous leishmaniasis caused by *Leishmania braziliensis* in Brazil: a randomized and controlled trial. PLoS Negl Trop Dis. 2010;4(12):e912.
DOI: 10.1371/journal.pntd.0000912.
42. Machado PRL, Ribeiro CS, Franca-Costa J, Dourado MEF, Trinconi CT, Yokoyama-Yasunaka JKU, *et al*. Tamoxifen and meglumine antimoniate combined therapy in cutaneous leishmaniasis patients: a randomized trial. Trop Med Int Health. 2018;23(9):936-942.
DOI: 10.1111/tmi.13119.
43. Machado PRL, Prates FVO, Boaventura V, Lago T, Guimaraes LH, Schriefer A, *et al*. A double-blind, randomized trial to evaluate miltefosine and topical granulocyte macrophage colony-stimulating factor in the treatment of cutaneous leishmaniasis caused by *Leishmania braziliensis* in Brazil. Clin Infect Dis. 2021;73(7):e2465-e2469.
DOI: 10.1093/cid/ciaa1337.
44. Martinez S, Marr JJ. Allopurinol in the treatment of American cutaneous leishmaniasis. N Engl J Med. 1992;326(11):741-744.
DOI: 10.1056/NEJM199203123261105.
45. Martinez S, Gonzalez M, Vernaza ME. Treatment of cutaneous leishmaniasis with allopurinol and stibogluconate. Clin Infect Dis. 1997;24(2):165-169.
DOI: 10.1093/clinids/24.2.165.
46. Mendes L, Guerra JO, Costa B, Silva ASD, Guerra M, Ortiz J, *et al*. Association of miltefosine with granulocyte and macrophage colony-stimulating factor (GM-CSF) in the treatment of cutaneous leishmaniasis in the Amazon region: a randomized and controlled trial. Int J Infect Dis. 2021;103:358-363.
DOI: 10.1016/j.ijid.2020.11.183.
47. Miranda-Verástegui C, Llanos-Cuentas A, Arevalo I, Ward BJ, Matlashewski G. Randomized, double-blind clinical trial of topical imiquimod 5% with parenteral meglumine antimoniate in the treatment of cutaneous leishmaniasis in Peru. Clin Infect Dis. 2005;40(10):1395-1403.
DOI: 10.1086/429238.
48. Miranda-Verástegui C, Tulliano G, Gyorkos TW, Calderon W, Rahme E, Ward B, *et al*. First-line therapy for human cutaneous leishmaniasis in Peru using the TLR7 agonist imiquimod in combination with pentavalent antimony. PLoS Negl Trop Dis. 2009;3(7):e491.
DOI: 10.1371/journal.pntd.0000491.
49. Mohebali M, Fotouhi A, Hooshmand B, Zarei Z, Akhoundi B, Rahnema A, *et al*. Comparison of miltefosine and meglumine antimoniate for the treatment of zoonotic cutaneous leishmaniasis (ZCL) by a randomized clinical trial in Iran. Acta Trop. 2007;103(1):33-40.
DOI: 10.1016/j.actatropica.2007.05.005.
50. Momeni AZ, Reiszadae MR, Aminjavaheri M. Treatment of cutaneous leishmaniasis with a combination of allopurinol and low-dose meglumine antimoniate. Int J Dermatol. 2002;41(7):441-443.
DOI: 10.1046/j.1365-4362.2002.01527.x.

51. Munir A, Janjua SA, Hussain I. Clinical efficacy of intramuscular meglumine antimoniate alone and in combination with intralesional meglumine antimoniate in the treatment of Old World cutaneous leishmaniasis. Acta Dermatovenerol Croat. 2008;16(2):60-64.
PMID: 18541100.
52. Nassiri-Kashani M, Firooz A, Khamesipour A, Mojtahed F, Nilforoushzadeh M, Hejazi H, *et al*. A randomized, double-blind, placebo-controlled clinical trial of itraconazole in the treatment of cutaneous leishmaniasis. J Eur Acad Dermatol Venereol. 2005;19(1):80-83.
DOI: 10.1111/j.1468-3083.2004.01133.x.
53. Navin TR, Arana BA, Arana FE, Berman JD, Chajon JF. Placebo-controlled clinical trial of sodium stibogluconate (Pentostam) versus ketoconazole for treating cutaneous leishmaniasis in Guatemala. J Infect Dis. 1992;165(3):528-534.
DOI: 10.1093/infdis/165.3.528.
54. Neves LO, Talhari AC, Gadelha EP, Silva Junior RM, Guerra JA, Ferreira LC, *et al*. A randomized clinical trial comparing meglumine antimoniate, pentamidine and amphotericin B for the treatment of cutaneous leishmaniasis by *Leishmania guyanensis*. An Bras Dermatol. 2011;86(6):1092-1101.
DOI: 10.1590/S0365-05962011000600005.
55. Ozgoztasi O, Baydar I. A randomized clinical trial of topical paromomycin versus oral ketoconazole for treating cutaneous leishmaniasis in Turkey. Int J Dermatol. 1997;36(1):61-63.
DOI: 10.1046/j.1365-4362.1997.00022.x.
56. Parhizkar AR, Sharafi M, Mansuri S, Hadibarhaghtalab M, Afrashteh S, Fatemian H, *et al*. Comparing the efficacy of fluconazole and cryotherapy versus cryotherapy alone on treating cutaneous leishmaniasis: a triple-blind randomized clinical trial. BMC Infect Dis. 2024;24(1):332.
DOI: 10.1186/s12879-024-09211-5.
57. Pennini SN, de Oliveira Guerra JA, Rebello PFB, Abtibol-Bernardino MR, de Castro LL, da Silva Balieiro AA, *et al*. Treatment of cutaneous leishmaniasis with a sequential scheme of pentamidine and tamoxifen in an area with a predominance of *Leishmania* (Viannia) *guyanensis*: a randomized, non-inferiority clinical trial. Trop Med Int Health. 2023;28(12):871-880.
DOI: 10.1111/tmi.13943.
58. Prates FV, Dourado ME, Silva SC, Schriefer A, Guimaraes LH, Brito MD, *et al*. Fluconazole in the treatment of cutaneous leishmaniasis caused by *Leishmania braziliensis*: a randomized controlled trial. Clin Infect Dis. 2017;64(1):67-71.
DOI: 10.1093/cid/ciw662.
59. Rahman A, Tahir M, Naveed T, Abdullah M, Qayyum N, Malik DH, *et al*. Comparison of meglumine antimoniate and miltefosine in cutaneous leishmaniasis. J Coll Physicians Surg Pak. 2023;33(12):1367-1371.
DOI: 10.29271/jcpsp.2023.12.1367.
60. Ribeiro CS, Franca RR, Silva JA, Silva SCD, Uliana SRB, Boaventura VS, *et al*. Cellular infiltrate in cutaneous leishmaniasis lesions and therapeutic outcome. An Bras Dermatol. 2021;96(5):544-550.
DOI: 10.1016/j.abd.2021.02.006.
61. Rubiano LC, Miranda MC, Muvdi Arenas S, Montero LM, Rodriguez-Barraquer I, Garcerant D, *et al*. Noninferiority of miltefosine versus meglumine antimoniate for cutaneous leishmaniasis in children. J Infect Dis. 2012;205(4):684-692.
DOI: 10.1093/infdis/jir816.
62. Sadeghian G, Nilforoushzadeh MA. Effect of combination therapy with systemic glucantime and pentoxifylline in the treatment of cutaneous leishmaniasis. Int J Dermatol. 2006;45(7):819-821.
DOI: 10.1111/j.1365-4632.2006.02867.x.
63. Saenz RE, Paz H, Berman JD. Efficacy of ketoconazole against *Leishmania braziliensis* panamensis cutaneous leishmaniasis. Am J Med. 1990;89(2):147-155.
DOI: 10.1016/0002-9343(90)90292-L.
64. Salmanpour R, Handjani F, Nouhpisheh MK. Comparative study of the efficacy of oral ketoconazole with intralesional meglumine antimoniate (Glucantime) for the treatment of cutaneous leishmaniasis. J Dermatolog Treat. 2001;12(3):159-162.
DOI: 10.1080/09546630152607899.
65. Sattar F, Akram S, Akhtar A, Almas U, Tahir R, Nawaz S. Efficacy of oral zinc sulphate compared with intralesional meglumine antimoniate injection (glucantime) in the treatment of cutaneous leishmaniasis. J Pak Assoc Dermatol. 2020;30(1):111-116.
DOI: 10.66344/jpad.v30i1.1338.
66. Shahzad MK, Tahir R, Khan S, Memon Q, Memon HS, Din GM. Efficacy of oral azithromycin for the treatment of cutaneous leishmaniasis. Pak J Med Health Sci. 2020;14(4):1962-1964.
DOI: Not available.
67. Shanehsaz SM, Ishkhanian S. A comparative study between the efficacy of oral cimetidine and low-dose systemic meglumine antimoniate (MA) with a standard dose of systemic MA in the treatment of cutaneous leishmaniasis. Int J Dermatol. 2015;54(7):834-838.

DOI: 10.1111/ijd.12709.
68. Soto J, Arana BA, Toledo J, Rizzo N, Vega JC, Diaz A, *et al*. Miltefosine for New World cutaneous leishmaniasis. Clin Infect Dis. 2004;38(9):1266-1272.
DOI: 10.1086/383321.
69. Soto-Mancipe J, Grogl M, Berman JD. Evaluation of pentamidine for the treatment of cutaneous leishmaniasis in Colombia. Clin Infect Dis. 1993;16(3):417-425.
DOI: 10.1093/clind/16.3.417.
70. Toledo Junior A, Daher AB, Amaral TA, Carvalho SF, Romero GA, Rabello A. Poor response to azithromycin in cutaneous leishmaniasis leading to a premature interruption of a multicentric phase III clinical trial in Brazil. Rev Soc Bras Med Trop. 2014;47(6):756-762.
DOI: 10.1590/0037-8682-0266-2014.
71. Vélez I, Lopez L, Sanchez X, Mestra L, Rojas C, Rodriguez E. Efficacy of miltefosine for the treatment of American cutaneous leishmaniasis. Am J Trop Med Hyg. 2010;83(2):351-356.
DOI: 10.4269/ajtmh.2010.10-0060.
72. Yazdanpanah MJ, Banihashemi M, Pezeshkpoor F, Khajedaluee M, Famili S, Tavakoli Rodi I, *et al*. Comparison of oral zinc sulfate with systemic meglumine antimoniate in the treatment of cutaneous leishmaniasis. Dermatol Res Pract. 2011;2011:269515.
DOI: 10.1155/2011/269515.
73. Zabolinejad N, Layegh P, Abbasi Shaye Z, Salehi M, Ghanizadeh S. Evaluating the effect of oral clarithromycin on acute cutaneous leishmaniasis lesions compared with systemic glucantime. J Dermatolog Treat. 2022;33(3):1418-1423.
DOI: 10.1080/09546634.2020.1825612.
74. Zerehsaz F, Salmanpour R, Handjani F, Ardehali S, Panjehshahin MR, Tabei SZ, *et al*. A double-blind randomized clinical trial of a topical herbal extract (Z-HE) versus systemic meglumine antimoniate for the treatment of cutaneous leishmaniasis in Iran. Int J Dermatol. 1999;38(8):610-612.
DOI: 10.1046/j.1365-4362.1999.00727.x.